

SCHOOL OF ENGINEERING
COCHIN UNIVERSITY OF SCIENCE AND TECHNOLOGY
B-TECH DEGREE SECOND INTERNAL EXAMINATION, FEB 2023
19-205-0302: ELECTRICAL TECHNOLOGY

Time: 2hrs

Max Marks : 50

CO 3: Understand the constructional details, operational principles, and performance characteristics of induction motors and alternators

CO 4: Learn about the generation, transmission, and distribution of electrical energy.

PART A

(Answer ALL questions)

(4x3=12)

1. Distinguish between two different types of synchronous machine [CO3]
2. Why synchronous motor is not self starting? How can it be made self starting? [CO3]
3. Derive corona effect. Explain methods to reduce it. [CO4]
4. What do you mean by load dispatching? [CO4]

PART B (Answer the questions)

5. (a) A 3 phase, 16 pole, star connected alternator has 240 stator slots with 8 conductors per slot and conductor of each phase are connected in series. The coil span is 144 electrical degree. Determine the phase and line emf if the machine speed is at 375 rpm and flux per pole is 6.1 mWb distributed sinusoidally in airgap [CO3] (7 marks)
- (b) Explain torque slip characteristics of three phase induction motor. [CO3] (5 marks)

OR

6. (a) Show that in induction motor, the maximum running torque is independent of rotor resistance. [CO3] (7 marks)

(b) Derive emf equation of alternator

[CO3] (5 marks)

7.

The data obtained on 100 kVA, 1100 V, 3 phase alternator is:

Open circuit test: Field current = 12.5 A dc, line voltage = 420 V ac

Short circuit test: Field current = 12.5 A dc, line current = rated value.

The ac resistance of the winding per phase is 0.5Ω . Calculate the voltage regulation of alternator at 0.8 pf lagging and 0.8 pf leading.

[CO3] (12 MARKS)

OR

8. A 6 pole , 50 Hz, 3 ϕ induction motor running on full load develops a useful torque of 160 Nm when the rotor emf makes 120 complete cycles per minute. The mechanical torque lost in friction is 10 Nm. The total stator loss is given to be 850 W. Calculate (i) Rotor slip (ii) Efficiency at full load (iii) Rotor Cu loss (iv) Shaft power output.

[CO3] (12 MARKS)

5. With neat diagram, suggest a suitable power plant which is always used as base load power plant in our country because of the abundant nature of fuel. Also mention its advantages and disadvantages. [CO4](14 marks)